

PROBLEM STATEMENT

Triple Composite Chain Rule

Find the derivative with respect to x of the function:

$$y = \log(\cos(e^{3x}))$$

STEP 1 প্রদত্ত ত্রি-স্তর যৌগিক ফাংশনটি উল্লেখ করি (Given Composite Function)

$$y = \log(\cos(e^{3x}))$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করি (Applying Differentiation Operator w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [\log(\cos(e^{3x}))]$$

STEP 3 সর্ববহিঃস্থ লগারিদম ফাংশনের জন্য চেইন রুল প্রয়োগ করি (Differentiating Outermost Logarithm)

$$\frac{dy}{dx} = \frac{1}{\cos(e^{3x})} \cdot \frac{d}{dx} [\cos(e^{3x})]$$

STEP 4 মধ্যবর্তী কোসাইন ফাংশনের উপর চেইন রুল প্রয়োগ করি (Differentiating Intermediate Cosine Function)

$$\frac{d}{dx} [\cos(e^{3x})] = -\sin(e^{3x}) \cdot \frac{d}{dx} (e^{3x}) \implies \frac{dy}{dx} = \frac{-\sin(e^{3x})}{\cos(e^{3x})} \cdot \frac{d}{dx} (e^{3x})$$

STEP 5 সর্বঅন্তরীণ সূচকীয় পদের অন্তরজ নির্ণয় করে ট্যানজেন্ট আকারে লিখি (Differentiating Innermost Exponential & Simplifying to Tangent)

$$\frac{d}{dx} (e^{3x}) = 3e^{3x} \implies \frac{dy}{dx} = -\tan(e^{3x}) \cdot 3e^{3x} = -3e^{3x} \tan(e^{3x})$$

FINAL ANSWER
$$\frac{dy}{dx} = -3e^{3x} \tan(e^{3x})$$

BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- Layer-by-Layer Decomposition:** Let $u = \cos v$ and $v = e^{3x}$. Then $y = \log u$.
$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx} = \left(\frac{1}{\cos v} \right) \cdot (-\sin v) \cdot (3e^{3x}) = -3e^{3x} \tan(e^{3x})$$
- Standard Trigonometric Identity:** $\frac{\sin \theta}{\cos \theta} = \tan \theta$.
- Exponential Chain Rule:** $\frac{d}{dx}[e^{kx}] = ke^{kx}$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- Negative Sign Dropped on Cosine Derivative:** $\frac{d}{dx}[\cos(\dots)] = -\sin(\dots)$. Omitting the negative sign is a critical loss-of-marks error.
- Missing Exponent Coefficient:** Forgetting to multiply by 3 from $\frac{d}{dx}(3x) = 3$, leaving $-e^{3x} \tan(e^{3x})$.
- Argument Contraction Mistake:** Writing $-3e^{3x} \tan x$ instead of $-3e^{3x} \tan(e^{3x})$. The argument e^{3x} must be preserved inside the tangent!

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- Logarithmic Positivity Requirement:** $\cos(e^{3x}) > 0 \implies e^{3x} \in (-\frac{\pi}{2} + 2k\pi, \frac{\pi}{2} + 2k\pi)$ for $k \geq 1, k \in \mathbb{Z}$ (or for $k = 0, e^{3x} \in (0, \pi/2)$).
- Principal Domain Region:** For $k = 0, 0 < e^{3x} < \frac{\pi}{2} \implies 3x < \ln(\pi/2) \implies x < \frac{1}{3} \ln(\frac{\pi}{2}) \approx 0.1505$.
- Differentiability:** Smooth and differentiable on all disjoint intervals where $\cos(e^{3x}) > 0$.

BOX 4: EXPLICIT STEP-BY-STEP SUBSTITUTION METHOD

- Let $v = e^{3x} \implies \frac{dv}{dx} = 3e^{3x}$
- Let $u = \cos v \implies \frac{du}{dv} = -\sin v$
- Then $y = \log u \implies \frac{dy}{du} = \frac{1}{u}$
- Combining all rates:
$$\frac{dy}{dx} = \frac{1}{\cos v} \cdot (-\sin v) \cdot 3e^{3x} = -\tan v \cdot 3e^{3x} = -3e^{3x} \tan(e^{3x})$$

BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- Step 1–2 (Function Setup & Chain Rule Strategy):** 1.0 Mark
- Step 3 (Outer Logarithmic Derivative):** 1.0 Mark
- Step 4 (Trig Derivative with Sign):** 1.0 Mark
- Step 5 (Exponential Inner Derivative & Tangent Form):** 1.0 Mark
- Total Allocation:** 4.0 Marks